# Coffee Futures & Option Valuation

QUANTITATIVE PRICING MEMO · 10 October 2026

## Executive summary

Using the supplied six-month market assumptions, the estimated fair futures price is $1.21814 per pound. A European call with a $1.25 strike is worth $0.06836 per pound using the spot-based Black-Scholes formula in the training guide; for a call specifically written on coffee futures, Black-76 yields $0.07119 per pound. Risk-neutral Monte Carlo provides an independent cross-check.

| VALUATION OUTPUT | USD PER LB | PURPOSE |
|---|---|---|
| Six-month futures - cost of carry | $1.21814 | Primary futures valuation |
| European call - spot Black-Scholes | $0.06836 | Training-guide convention |
| European call on futures - Black-76 | $0.07119 | Preferred futures-option convention |
| European call - Monte Carlo | $0.07141 | Simulation validation |

## Provided market inputs

| INPUT | VALUE | INPUT | VALUE |
|---|---|---|---|
| Spot coffee price | $1.20 / lb | Risk-free rate | 2.00% p.a. |
| Storage cost rate | 1.00% p.a. | Maturity | 0.50 years |
| Call strike | $1.25 / lb | Volatility | 25.00% p.a. |

## 1 | Cost-of-carry futures pricing

$F = S \times \exp[(r + d) \times T] = 1.20 \times \exp[(0.02 + 0.01) \times 0.50] = \$1.21813568$ / lb.

Interpretation: financing and physical storage increase the six-month no-arbitrage carrying cost by approximately $0.01814 per pound. This simplified model assumes continuously compounded rates and zero convenience yield; it excludes transaction costs, delivery-grade differences, taxes and market frictions.

## 2 | European call option pricing

Spot Black-Scholes (as in the training guide): $C = S\,N(d1) - K \exp(-rT)\,N(d2)$. With S = $1.20, K = $1.25, r = 2%, T = 0.50, and sigma = 25%, the call is $0.06836 / lb.

Black-76 for an option on a futures contract: $C = \exp(-rT)[F\,N(h1) - K\,N(h2)]$, where $h1 = [\ln(F/K) + 0.5\,\text{sigma}^2 T] / [\text{sigma}\,\sqrt{T}]$ and $h2 = h1 - \text{sigma}\,\sqrt{T}$. Substituting the cost-of-carry future gives $0.07119 / lb.

## 3 | Risk-neutral Monte Carlo cross-check

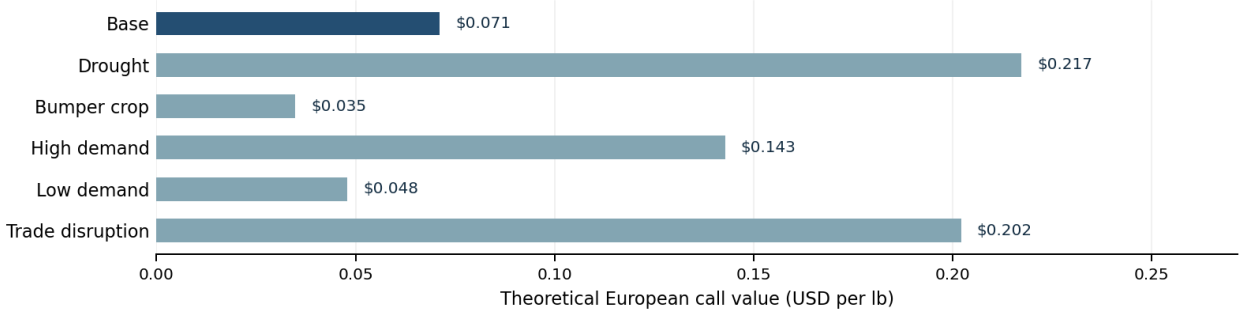
200,000 independent terminal-price simulations, with random seed 42, use $S(T) = S(0) \exp[(r + d - 0.5\,\text{sigma}^2)T + \text{sigma}\,\sqrt{T} \times Z]$, $Z \sim N(0,1)$. With zero convenience yield, the simulated terminal mean should approximate F; the discounted mean of $\max[S(T) - K, 0]$ estimates the European call value.

Results: mean simulated terminal coffee price = $1.21809 / lb (theoretical $1.21814); simulated call value = $0.07141 / lb, standard error $0.00029 / lb (approximate 95% Monte Carlo error band: $0.07084 to $0.07197).

## 4 | Supply, demand, weather and geopolitical scenarios

The table below is a scenario sensitivity analysis, not an empirical forecast. Spot-price shocks and volatility assumptions are hypothetical analyst choices; the task provides no observed estimates for these shocks. Carry rates and maturity remain fixed. Each option value is calculated with Black-76.

| SCENARIO | SPOT | VOL. | FUTURES | CALL |
|---|---|---|---|---|
| Base case | $1.200 | 25% | $1.218 | $0.071 |
| Severe drought / supply loss | $1.380 | 35% | $1.401 | $0.217 |
| Bumper harvest | $1.080 | 28% | $1.096 | $0.035 |
| Stronger global demand | $1.296 | 30% | $1.316 | $0.143 |
| Demand slowdown | $1.104 | 30% | $1.121 | $0.048 |
| Trade / geopolitical disruption | $1.344 | 38% | $1.364 | $0.202 |




### How to interpret the drivers

**Supply and weather:** Drought, frost, pests and shipping disruption may tighten available coffee supply, raising spot and futures prices. A bumper crop generally has the opposite effect.

**Demand:** Stronger consumption can lift prices and the expected call payoff; weaker consumption can reduce them, all else equal.

**Geopolitics and FX:** Export restrictions, political instability, freight costs and producer-currency movements can alter effective availability and risk premia.

**Option sensitivity:** Higher underlying futures prices and greater expected volatility usually increase a European call value, subject to other assumptions.

### Pricing decision and risk controls

Use $1.21814/lb as the stylized six-month futures fair-value estimate. For the stated European call on this future, use approximately $0.07119/lb as the preferred theoretical premium; the Monte Carlo estimate provides a consistent validation. Actual tradable prices require the quoted futures curve, option conventions, current implied volatility, contract specifications, transaction costs and calibration to market data.

Risk controls: monitor price and volatility sensitivities (delta/vega), exposure limits, stress losses, basis risk, margin and liquidity. Refresh assumptions after crop reports, demand revisions, weather alerts or policy changes.

### Methodological notes

*The call is assumed to be European and cash-settled at maturity. Annual rates are continuously compounded. Coffee prices and premiums are expressed per pound; contract-level USD amounts require the exchange contract size. The task inputs are instructional assumptions, not verified live market quotes. Simulations under risk-neutral pricing are valuation tools, not predictive forecasts of real-world expected returns.*

Reference: Citi / Forage "Pricing Commodities - Coffee" task data and the accompanying "Step by step guide" (provided with the training material).